

Introduction to Integrals*

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1 Definition

Let $y = f(x)$ be a function defined on an interval $[a, b]$. How do we find the area under the curve?

First we can partition $[a, b]$ into n equal sub-intervals. Each sub-interval has the length $\Delta x = \frac{b-a}{n}$. a and b are the endpoints of the sub-intervals. So,

$$a = x_0 < x_1 < x_2 < \dots < x_n = b \text{ on the } i^{th} \text{ sub-interval } [x_{i-1}, x_i]$$

Next, we can draw a rectangle whose height is $f(x)$ on the sub-interval. There area of the rectangle is:

$$f(x_n^*)(x_n - x_{n-1}) = f(x_i^*)(\Delta x)$$

So, the area under the graph on $[a, b]$ is approximately the sum of the areas of the rectangles we drew:

$$\text{Area} \approx \sum_{i=1}^n f(x_i^*)(\Delta x)$$

This is called the Riemann Sum.

*Definitions and Examples from Dr. Donald White's Lecture

As $n \rightarrow \infty$, the widths of the sub intervals go to zero and our Riemann Sum approaches the exact area. So we can take the limit of the Riemann Sum to find the exact area:

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)(\Delta x)$$

This is equal to the definite integral of $f(x)$ on $[a, b]$:

$$\int_a^b f(x) dx$$

If the limit exists, the integral exists and we say $f(x)$ is integrable on $[a, b]$. a is the lower limit of integration, and b is the upper limit. dx is the differential, it tells the variable we are integrating with respect to. So, in this case we would say "the integral from a to b of $f(x)$."

To evaluate integrals we take the anti-derivative of the function we are integrating. If $F(x)$ is a function and $f(x) = F'(x)$, then $F(x)$ is the anti-derivative of $f(x)$. All anti-derivatives are of the form $F(x) + C$. C is a constant. So,

$$\frac{d}{dx}(F(x) + C) = \frac{d}{dx}F'(x) + 0 = F'(x)$$

In other words, we are given a derivative and are trying to find the original function. We are "undoing" derivatives. We add $+C$ because there is no way to recover any constants from the original equation, given the derivative only.

To do an anti-derivative, we add 1 to the exponent, then divide the base by the new exponent. Then we add C . So the anti-derivative of x^n is,

$$\frac{1}{n+1} x^{n+1} + C \text{ for all } n \neq -1$$

2 Properties Anti-Derivatives

1. $\frac{d}{dx} cF(x) = c \frac{d}{dx} F(x)$

That is, if $F(x)$ is an anti-derivative of $f(x)$, then $cF(x)$ is an anti-derivative of $cf(x)$.

2. If $F'(x) = f(x)$ and $G'(x) = g(x)$, then $\frac{d}{dx} (F(x) + G(x)) = F'(x) + G'(x) = f(x) + g(x)$

That is, $F'(x) + G'(x)$ is an anti-derivative of $f(x) + g(x)$

3. The anti-derivative of a quotient is not the quotient of the anti-derivatives
4. The anti-derivative of a product is not the product of the anti-derivatives.

3 Properties of Integrals

1. $\int_a^a f(x) dx = 0$
2. $\int_b^a f(x) dx = - \int_a^b f(x) dx$
3. If C is constant then $\int_a^b C dx = C(b - a)$
4. $\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$
5. $\int_a^b cf(x) dx = c \int_a^b f(x) dx$
6. $\int_a^b (f(x)g(x)) dx \neq (\int_a^b f(x) dx)(\int_a^b g(x) dx)$

There are two kinds of integrals, definite and indefinite. Definite integrals are evaluated on $[a, b]$ like in the ones above. Unlike anti-derivatives and indefinite integrals, definite integrals do not get a $+C$, since when we evaluate the integral on $[a, b]$ we always get a $+C$ and a $-C$, so they cancel. Indefinite integrals are not evaluated on a certain interval, and therefore always get $+C$.

An example of a indefinite integral is:

$$\int_a^b f(x) dx$$

And a indefinite integral:

$$\int f(x) dx$$

4 Examples

1. Evaluate $\int 3x^3 dx$

Our first step is to find the anti-derivative of $3x^3$. To do this, we first add one to the exponent.

$$3x^4$$

Now we will divide the base by the value of our new exponent. Then we add C .

$$\frac{3}{4}x^4 + C$$

Since this is an indefinite integral and is not evaluated between any limits, we are done and have arrived at our answer.

2. Evaluate $\int_3^4 3x^3 dx$

Our first step is to find the anti-derivative of $3x^3$:

$$\frac{3}{4}x^4 + C$$

We then evaluate the function 3 to 4.

$$\frac{3}{4}x^4 \Big|_3^4 =$$

$$\frac{3}{4}(4)^4 - \frac{3}{4}(3)^4 =$$

$$\frac{525}{4}$$